Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 1 of 8 · Question 5 part b · 7 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Question number					
5	(a)	(i)	• prevent contamination/no mixing with water; • because rainwater will flow away from well;		2
		(ii)	allow urine/liquid waste, to soak away/absorbed by soil;		1
		(iii)	• prevent smell; • covered to prevent entry of flies/rats; • as these carry bacteria/pathogens; • and spread disease; • long handle to keep hands away from pit entrance/cover faeces; • more likely to be bacteria/pathogens in that area;		5
	(b)		• any seven from • equal volumes of two solutions; • same concentrations; • wash concrete slab with one solution; • leave for specified time; • use sterile transfer loop or equivalent; • transfer from slab to Petri dish/nutrient agar; • incubate for specified time; • count colonies/measure inhibition zones; • repeat with second solution; • compare results;		7

Total 15

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 1 of 8 · Question 5 part b · 7 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 2 of 8 · Question 4 part b.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

4	(a)	(i)	A (psychrophiles); B is incorrect as the temperature of a fridge is not between 12 and 45 degrees C is incorrect as the temperature range is too high for a fridge D is incorrect as the temperature range is too high for a fridge		1
		(ii)	increase in growth/maximum growth; (near) optimum/best temperature;	Allow optimum growth.	1 1
	(b)	(i)	measure diameter/area of inhibition/of the clear zone/area of no bacterial growth; greater the diameter, the more effective the antibiotic;	Allow observe size of the clear zone.	2
		(ii)	wash hands/work surface after carrying out procedure; reduces transfer of bacteria; Heat inoculating loop/spreader; To kill bacteria/to prevent contamination; do not open dish following incubation/seal the Petri dish (prior to incubation); reduces risk of bacteria being transferred to student/infection; do not incubate over 25°C; prevents growth of pathogenic bacteria; Sterilise/clean all equipment before/after use; Reduces cross contamination; Seal petri dish/do not open lid of petri dish too far/reduce exposure of contents of dish to air; Reduces contamination by airborne bacteria/to prevent bacteria escaping;	Max 4 if no explanation given.	6 max

Total question 4 = 11 marks

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 2 of 8 · Question 4 part b.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 3 of 8 · Question 4 part b.ii · 6 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

4	(a)	(i)	A (psychrophiles); B is incorrect as the temperature of a fridge is not between 12 and 45 degrees C is incorrect as the temperature range is too high for a fridge D is incorrect as the temperature range is too high for a fridge	1
		(ii)	increase in growth/maximum growth; (near) optimum/best temperature;	Allow optimum growth.	1 1
	(b)	(i)	measure diameter/area of inhibition/of the clear zone/area of no bacterial growth; greater the diameter, the more effective the antibiotic;	Allow observe size of the clear zone.	2
		(ii)	wash hands/work surface after carrying out procedure; reduces transfer of bacteria; Heat inoculating loop/spreader; To kill bacteria/to prevent contamination; do not open dish following incubation/seal the Petri dish (prior to incubation); reduces risk of bacteria being transferred to student/infection; do not incubate over 25°C; prevents growth of pathogenic bacteria; Sterilise/clean all equipment before/after use; Reduces cross contamination; Seal petri dish/do not open lid of petri dish too far/reduce exposure of contents of dish to air; Reduces contamination by airborne bacteria/to prevent bacteria escaping;	Max 4 if no explanation given.	6 max

Total question 4 = 11 marks

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 3 of 8 · Question 4 part b.ii · 6 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 4 of 8 · Question 6 part b.ii · 6 marks

2020 JAN · Paper 02 · find part b.ii in this mark scheme extract

Number				
6	(a) (i)	• more males than females die (from *S.aureus* and MRSA)(1) • number of deaths from both *S.aureus* and MRSA increase with age(1)	Allow other valid conclusions e.g more deaths from both microorganisms in males/females aged 85 and over	2
	(ii)	416.6 ÷ 1.8(1) 231.4:1 (1)	Allow 231:1 Allow full marks for correct final answer	2
	(iii)	41.2 + 21.9 = 63.1(1) 53 × 63.1 = 3344(1)	Allow 3344.3 Allow full marks for correct final answer	2
	(iv)	Any two from: • MRSA is resistant (1) • to (many) antibiotics(1) • (MRSA infection) difficult to treat(1) • reference to MRSA infection being (highly) contagious/out of control/easily transferred(1)		2
	(b) (i)	mircroorganisms/bacteria/fungi	Ignore viruses Ignore named antibiotics	1
	(ii)	Any six from: • sterilise Petri dishes/agar/nutrient broth(1) • obtain sample(1) • sterile inoculating loop/sterile swab(1) • pass neck of flask containing culture through a flame(1) • smear culture/sample/swab onto agar(1) • reference to adding antibiotics (to agar)(1) • seal Petri dishes/cover with lid(1) • incubate(1) • measure area of clear zone around antibiotic discs(1)	Allow description of how loop/swab is sterilised	6

Total for Question 6 = 15 marks

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 5 of 8 · Question 7 part c · 2 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

number				
7 (a)	(i)	• temperature of incubation; • volume of milk; • composition of nutrients in agar/type of agar;		3
	(ii)	size of bacterial growth;		1
(b)	(i)	$\frac{7.1}{5} = 1.42$; $\frac{36.5}{5} = 7.3$;	One mark for each correct answer	2
	(ii)	• more bacteria in B; • at start of investigation; • because extra added to tube; • some bacteria in A/milk which grew;	Allow more growth in B/growth increases more quickly in B Allow reference to contamination e.g.from air	4
(c)		• transfer loop sterilised/heated in flame; • transfer performed quickly;	Allow reference to sterilised in context / sterile technique / sterilise equipment / named equipment	2
(d)		• repeat investigation; • using other organisms;		2

Total 14 marks

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 5 of 8 · Question 7 part c · 2 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 6 of 8 · Question 6 part b · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

6 (a)	(type of) antibacterial cleaner;		1
(b)	Any two from the following • use sterilised equipment/named equipment; • heat the inoculating loop (in a Bunsen flame); • do not open the lid of the Petri dish too much; • seal the lid of the Petri dish; (with tape)	Named equipment must be hardware	2
(c)	• to prevent growth/reproduction; • of pathogen/harmful microorganism;		2
(d)	it is a control;		1
			Total 6

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 6 of 8 · Question 6 part b · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 7 of 8 · Question 5 part a.ii · 2 marks

2024 JUN · Paper 02 · find part a.ii in this mark scheme extract

Number				
5	(a) (i)	**C** the type of antibiotic added The following are incorrect because: A the mass of bacteria added is irrelevant B the size of the inhibition zone is the dependent variable D the mass of agar in the Petri dish is irrelevant		(1)
	(ii)	Any two from: • sterilise equipment/named equipment/disinfect/swab desk; • use of sterile agar; • reference to use of lid on Petri dish/ minimising the time the lid is removed; • Flame from Bunsen burner:		(2)
	(iii)	(antibiotic) Q; greater inhibition zone /greater clear zone/ more bacteria killed;		(2)
	(iv)	Any four from: • mutation; • (resistant) bacteria survive; • reproduce; • pass on resistant gene/allele; • Increase of resistant gene/allele in the population/ increase in allele frequency (over many generations)		(4)
	(b)	• (bacteria/fungi/decomposers) release enzymes/named enzyme • breakdown dead/organic matter/ named organic matter; • release nutrients/named nutrients into the soil/environment;	Ignore carbon dioxide Ignore fertiliser	(2)

Total for Question 5 = 11 marks

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 8 of 8 · Question 6 part c · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

6 (a)	(type of) antibacterial cleaner;		1
(b)	Any two from the following • use sterilised equipment/named equipment; • heat the inoculating loop (in a Bunsen flame); • do not open the lid of the Petri dish too much; • seal the lid of the Petri dish; (with tape)	Named equipment must be hardware	2
(c)	• to prevent growth/reproduction; • of pathogen/harmful microorganism;		2
(d)	it is a control;		1
			Total 6

Type MS · Use aseptic technique to test antibacterial agents on agar

Answer 8 of 8 · Question 6 part c · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks